

Flipkart APM ’29 | Problem Statement

Problem Statement

It’s 2018 & the Battle of Wakanda has just concluded. Thanos has snapped his fingers after completing the Infinity Gauntlet (gaining possession of all the Infinity Stones) & 50% of the population has suddenly been eliminated. Build a digital product for this new world **as on the day after the Snap**.

Individual’s Details

[Your Full Name]

[Your Enrollment Number (Institute ID)]

[Your Email]

[Your Phone Number]



Khairiyat (ख़ैरियत): no unresolved household without an accepted next step

1. What problem will Khairiyat solve?

The moment. The morning after half of all people vanished, the secretary of a (fictional) 160-flat Gurugram society must answer two questions for every flat: has anyone reached this household, and does anyone here need help? The loss is uneven: one home has lost a carer, another a parent, another no one. Silent flats prove nothing: a flat may be empty, or a child or bedridden adult may be unable to answer. Guards and volunteers are fewer too, and reports arrive scattered and contradictory.

The failure I target is follow-up that silently disappears: a flat marked OK on a child's word, a volunteer who never said yes, a missed deadline, a service called but never committed, a rumour treated as fact. In 2018 societies coordinated through WhatsApp groups, which carry messages but no per-flat status [1] and spread rumours widely enough that WhatsApp limited forwarding in India in July 2018 [2], and through spreadsheets, which hold status but do not enforce follow-through.

Why this problem. I have lived in apartment societies like this since birth, so the guard room, intercom, RWA and residents' WhatsApp group that Khairiyat builds on are the setting I understand best. I also scored six Delhi-NCR problems (Table 1). Hospital transfers scored highest but depend on bed capacity and government ownership; society roll-call scored lower on severity and evidence, but highest where software itself closes the gap and an RWA can adopt it on day one. Its main risk, a well-run spreadsheet being enough, is what my validation tests.

Problem	Urgency (10)	Severity (15)	Clear user (10)	Evidence (15)	Software matters (15)	Day-one (10)	Adoption (10)	Distinct (10)	Prototype (5)	Total
Hospital transfers	5	5	4	5	4	3	3	3	5	83
Society roll-call (chosen)	**4**	**3**	**5**	**2**	**5**	**5**	**4**	**3**	**5**	**77**
Infant feeding	5	5	4	4	2	2	3	5	4	75
Construction workers	3	4	4	5	3	2	2	4	3	69
Water tankers	4	4	4	4	2	3	3	2	3	65
Older adults alone	4	4	4	4	2	3	2	2	4	64

Table 1. Ratings 1 to 5, weighted to 100 (weights in brackets); my judgement, not measurement.

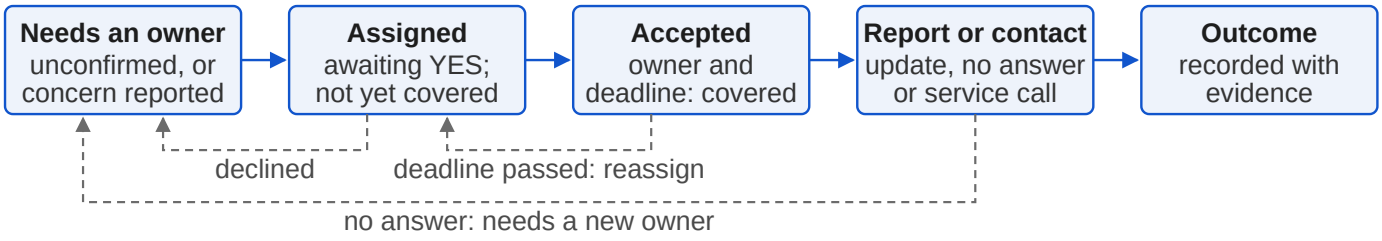
2. Who am I building it for?

Primary user: the roll-call coordinator of one large Delhi-NCR housing society, an RWA office-bearer or volunteer working from the society office on one laptop or phone: not technical, under pressure, possibly bereaved. **Affected but not using it:** wardens, guards and volunteers who receive task messages; residents and relatives who send information and receive totals-only updates. **Initial scope:** an organised high-rise society with a flat register, guard room and intercom (assumptions), which favours better-off societies.

3. What is Khairiyat, and how does it solve the problem?

I propose a single-device coordination desk built on one rule: **every unresolved item has a next step that a named person has accepted, with a deadline, and nothing closes without evidence** (Figure 1).

- **Two separate records per unit** (lower lane). Confirmed never means everyone is safe.
- **Accepted ownership.** Unaccepted work is not covered; a NO returns units to the queue; overdue work keeps its owner until extended with a reason or reassigned.
- **No invented evidence.** Strict bulk replies: a missing channel, free text or a typo is flagged, never guessed. Corrections are added, never overwritten.
- **Safe outputs.** Volunteer messages forbid forcing entry and say to call emergency services directly; Khairiyat never calls or posts anything. Society updates are totals only.



Two separate records per unit; closing one never closes the other

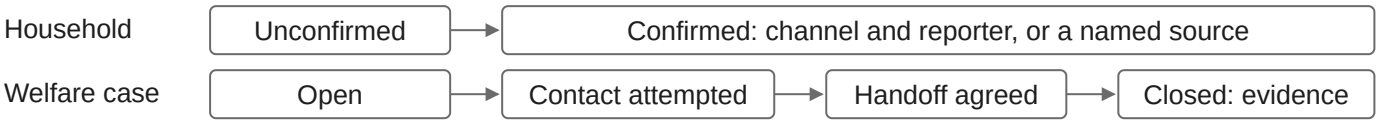


Figure 1. The next-step loop, and the two records each unit carries.

Why not WhatsApp plus a well-run Google Sheet? WhatsApp stays the messaging channel. A Sheet can hold the same columns but does not block a confirmation without a basis, reopen a gap when a volunteer declines or a deadline passes, or keep history instead of overwriting cells. It is better for simultaneous editors and familiarity; whether Khairiyat's guardrails outweigh that is what I must test.

The prototype (one HTML file, fictional data) has an 18-step guided walkthrough that ends honestly: 129 of 160 households confirmed, the other 31 with accepted owners and deadlines, and two welfare cases still open with their reasons.

4. How will I know it is working?

Primary metric: next-step coverage. It shows whether follow-up is under control, not that anyone is safe, so it always appears beside the unconfirmed and open-case counts (Table 2).

Measure	Definition	What it does not show	How collected
Next-step coverage (primary)	Share of unresolved units whose next step is accepted and not overdue; time to first reach 100%	That anyone is safe or any case is resolved	Event log, automatically
Households confirmed	Confirmed units at fixed checkpoints	That everyone in a unit is safe	Event log
Awaiting acceptance; overdue	Counts at the same checkpoints	Whether the work was done well	Event log
Time to outcome	Median time from case opening to outcome with evidence	Whether the outcome was right	Event log
Incorrect closures (guardrail: 0)	Confirmations or closures contradicting the scenario's ground truth	Real-world error rates	Comparison test
Privacy exposures (guardrail: 0)	A unit number, name or concern in a group message	Leaks outside the app	Review of copied messages

Table 2. Success measures; targets wait for a baseline from the comparison test.

Validation I propose (nothing has been run). (1) Think-aloud sessions with about five RWA volunteers. (2) Coordinator pairs run the same scripted scenario, with ten identical test cases, in Khairiyat and in a well-run Google Sheet. I continue only if Khairiyat shows fewer incorrect closures or fewer unowned or overdue tasks without being materially slower; otherwise I publish the Sheet template and printable floor sheets instead. (3) A drill in one consenting tower.

5. Risks and limits

- It tracks units, not people, and is not a missing-person system, emergency service or triage tool. Full coverage does not mean the emergency is over, and assigning everything to the desk can game it.
- One device, no accounts or encryption. Health details are sensitive personal data under India's 2011 IT rules [3], so a real version needs authentication, encryption and retention rules.

Appendix: references, and what is fact, assumption or hypothesis

A1. References

1. **WhatsApp groups in 2018.** Group size raised to 256 members in February 2016: Android Police, “WhatsApp group chats can now have up to 256 members”, 5 February 2016, androidpolice.com/2016/02/05/whatsapp-group-chats-can-now-have-up-to-256-members/. In-chat polls arrived only in November 2022: WhatsApp blog, “Communities now available”, blog.whatsapp.com/communities-now-available. Supports: a 2018 group chat could pass messages but had no structured per-flat responses.
2. **Forwarding limit in India, July 2018.** WhatsApp blog, “More changes to forwarding”, 19 July 2018, blog.whatsapp.com/more-changes-to-forwarding; Business Today, “To curb fake news, WhatsApp limits message forwarding to 5 users in India”, 21 July 2018, businesstoday.in/latest/economy-politics/story/fake-news-whatsapp-limits-message-forwarding-5-chats-facebook-108281-2018-07-21. Supports: WhatsApp tested a five-chat forwarding limit in India after mob violence linked to false messages; the same report gives WhatsApp’s user base in India as 200 million.
3. **Sensitive personal data.** Information Technology (Reasonable Security Practices and Procedures and Sensitive Personal Data or Information) Rules, 2011, G.S.R. 313(E), 11 April 2011, Rule 3 (includes physical, physiological and mental health condition, and medical records and history). Copy at prsindia.org/files/bills_acts/bills_parliament/2011/IT_Rules_2011.pdf.

The film does not state the Snap’s month in 2018, so no part of the product depends on a feature or rule that changed during 2018.

A2. Facts, assumptions and hypotheses

Facts with sources: the three items above.

Fictional premise and scenario assumptions: half of all people vanished at once and unevenly; the society, its 160 flats, about 550 former residents, every person and every message are invented; the society has a flat register, a guard room with intercom, a laptop or phone at the desk, and enough willing volunteers; mobile networks are congested but checks inside the society do not depend on them.

Untested product hypotheses: a guided workflow produces fewer incorrect closures and fewer unowned or overdue follow-ups than a well-run spreadsheet, without being materially slower; coordinators can learn it under pressure; volunteers will reply YES or NO to a structured message.

Problem-selection scores (out of 100, my own weighted judgement): hospital transfers 83; society roll-call 77; infant feeding 75; construction workers 69; water tankers 65; older adults alone 64. Criteria and weights: urgency (10), severity (15), user specificity (10), evidence (15), software makes the difference (15), day-one feasibility (10), adoption (10), differentiation (10), prototype shows the workflow (5).

No research is claimed. No interviews, surveys, usability sessions or pilots have been conducted. The problem-selection scores are my own judgement against stated criteria, not measurements.

A3. How to see the prototype

Unzip the prototype folder and open **index.html** in a browser; no installation, server or internet connection is needed. Choose **More → Reset demo (empty register)** and follow the walkthrough panel from step 1 to step 18. The final step’s handover summary shows the end state described in section 3.